# ENVIRONMENTAL POLLUTION
## (Air, Noise and Nuclear Pollution)

## INTRODUCTION

E.P. Odum (1971) defined pollution as an alteration in the physico-chemical and biological nature of air, water and soil that ultimately affects the whole environment. It also specifies its hazardous impacts on living organisms (both flora and fauna), other environmental systems and non-living material.

Pollution is caused by anthropogenic (man-made) or natural activities. Examples of natural pollutions are volcanic eruptions, forest fires, floods caused etc. Incomplete technology lacking close integrated systems is the main cause of man-made pollution.

Today, the problem of pollution has become a major challenge to scientists, environmentalists and humanists as the pollution of various components has gone to such an extent that we are unable to breath fresh air, drink fresh water and eat pure food. If man has to survive, he has to fight and overcome this gigantic problem before it swallows him and his very existence.

Sources of Pollution

Pollution may be caused by several sources depending upon the nature of pollutants:

1. **Solid Wastes as a Source of Pollution:** Solid Wastes may be domestic or industrial in nature. Various solid wastes can be categorised as follows:
   **(i)** Industrial wastes, e.g. particulate wastes from various industries such as glass fragments, leather pieces, rubber pieces etc.

**(ii)** Domestic Wastes, e.g. garbage of kitchen, slaughter houses etc. These may be combustible (such as leaves, twigs, papers etc.) or non-combustible (such as crockery, plastics, glass etc.) in nature.

**(iii)** Sewage, e.g. human and animal excreta, domestic effluents, detergents etc. (solid faecal is called sludge)

**(iv)** Agricultural Wastes, e.g. plant and animal residues, broken twigs, wood fractions, fruits, pesticides, fertilizers etc.

2. **Liquid Wastes as a Source of Pollution:** Industrial effluents and domestic wastes in the form of liquid are the major sources of water and soil pollution. Industrial pollution may leach the lethal magnitude. Liquid discharges from chemical factories, refineries, breweries, tanneries etc. contain acids, alkalies, oil and dissolved heavy metals which enter the river water and adversely affect the aquatic life and impair its self-purification system.

   Liquid wastes from domestic sources may be of inorganic or organic nature. Inorganic liquid wastes include soap water and detergent water from bathing and washing clothes whereas organic wastes include kitchen garbage, faecal water, urine etc.

   Run off from agricultural fields carry residual fertilizers, pesticides, biocides etc. which enter the water streams and harm the aquatic life.

3. **Gaseous Wastes as a Source of Pollution:** The common gaseous pollutants like carbon monoxide, sulphur oxides, nitrogen oxides, hydrogen sulphide etc. are frequently released from various industries and automobile exhausts. Pollution resulting from gaseous wastes is the one of the most dangerous and lethal type. Life on earth is dependent upon the air we breathe and if this source of life is contaminated and polluted by lethal gaseous discharges, the very existence of all living organisms is threatened.

4. **Energy Wastes as a Source of Pollution:** A significant addition to the sources of pollution are the invisible pollutants i.e. pollutants without mass or weight and invisible to the eyes. Examples of such pollutants are heat and radioactive emissions. Radioactive emissions are most hazardous, and their cumulative effects are far reaching and damaging to the genetic makeup of living organisms.

5. **Noise as a Source of Pollution:** Unwanted sound or noise above the particular level in the atmosphere is an important pollutant. Indiscriminate and continuous use of radios, traffic horns, public broadcasting systems etc. are common sources of noise pollution. Aeroplanes and supersonic jets also produce noise of high intensity which may sometimes rupture the ear drum and cause irreparable damage to the brain.

## 2. Kinds of Pollution

Pollution has been classified in various ways based on different factors such as on the basis of:

1. **Nature of Pollutants**

   Two categories are recognised on the basis of degradability of pollutants:

   **(i)** **Biodegradable:** These are those substances which can be decomposed naturally in the presence of microorganisms. For example, any organic waste, leaf litter etc.

   **(ii)** **Non-biodegradable:** These are those substances which cannot be decomposed or break down into simpler substances by any natural process. For example, plastic waste, metallurgical waste etc.

2. **Components of Environment**

   Pollution has been classified on the basis of particular component of the environment being polluted, such as:

   - Air Pollution
   - Water Pollution
   - Soil Pollution
   - Radioactive Pollution
   - Noise Pollution
   - Thermal Pollution

3. **Management of Environmental Pollution**

At present what is required is not so much the ability of human beings to conquer the nature, rather there is a need of a balanced and harmonious collaboration with its forces. The ultimate goal of environmentalists should be to manage the environment in such a way that it can contribute to man's happiness, health and enjoyment and to improve the quality of human life.

The various activities of civilisation invariably interact with the environment predicting various forms of environmental problems. These problems may be local, national or international in nature and must be managed accordingly. Following are few suggestions to combat pollution:

1. Environmental education must be made compulsory in all education systems, from primary to university levels.
2. Awareness about general environmental issues should be circulated at large scale through media services.
3. Formulation of quality standards should be implemented and breaking of these standards should be a punishable offence.
4. Sustainable approach towards environment should be made through community participation.

5. Relocation of industrial units, airports etc at the countryside to reduce the risk of air pollution.
6. Private and public enterprises should adopt pollution mitigation plans through laws and policies.
7. Nuclear testing should restricted to the minimum by international agreements.

# Air Pollution

It is defined as the pollution caused by anthropogenic and natural agents that degrades the quality of air and having impact on living and non-living systems. The essential component of our life, "air", resides in one of the spheres of the environment, called as "Atmosphere". The envelope of air surrounds the earth's surface is called as "atmosphere". It consists of both major and minor components which includes gases, particles, water vapour and biological particles. It makes up nearly 80% of man's daily intake by weight. Humans breathe almost 22,000 times per day, taking in almost 16kg of air. So, quality of air that we breathe is very important for our sustenance of life, otherwise it leads to loss of lives and produce disturbance in ecological systems. The atmosphere has two natural detergents that have the capacity to remove the pollutants from an air parcel: high mixing height plus high wind speed which favours high dispersion and high precipitation in the form of rainfall. Rainfall act as the best scavenging agent for removal of air pollutants.

## 1. Categorization of Air Pollutants

Agents or carriers which cause air pollution are known as air pollutants. Air Pollutants are divided on the basis of their origin, composition and kind of matter.

*A. Origin:*

i. Primary air pollutants are those which have their own origin and have direct emission into the atmosphere. For example, oxides of nitrogen, oxides of sulphur, oxides of carbon, volatile organic compounds and particulate matter (dust particles).

ii. Secondary air pollutants are those which are derived from primary air pollutants. For example, $O_3$, PAN etc.

*B. Chemical Composition:*

i. Organic compounds: These are those which contain carbon and hydrogen units. For example, aldehyde (formaldehyde) and ketone (acetone).

ii. Inorganic compounds: These are those which are having mixed formation of compounds and doesn't contain carbon and hydrogen units. For example, carbonates, nitrogen oxide etc.

## 2. Types of Air Pollution

Indoor air pollution: This type of air pollution has mainly the anthropogenic source. Domestic activities like cooking fuel burning, coal burning, crop residue burning etc. Mostly household women are severely affected due to emissions of indoor air pollutants. Approximately 2.5 Billion people, mostly in developing nations are exposed to high levels of indoor air pollution and consequently suffer.

Outdoor air pollution: Vehicular and industrial emissions are major sources of outdoor air pollution. Overpopulation growth and industrialization are also the major indirect contributors. Outdoor air pollution affects over 1,100 million people, predominantly in cities.

## 3. Sources of Air Pollution

A. *Natural Sources:*

The natural sources can be thunderstorms, emissions of gases and particulates from forest fires, methane emissions from marshy lands, bioaerosols from pollen grains, volatile organic compounds emissions from trees and plant species, volcanic eruptions generates sulphur dioxides emissions and decomposition of organic matter emits carbon dioxide and methane.

B. *Anthropogenic Sources:*

Along with natural pollutants, there are pollutants of anthropogenic origin too. The main anthropogenic sources are gasoline exhaust emissions, industrial emissions, mining activities, cooking fuels, construction works, fireworks etc. The details of each emission source categories are as follows:

i. Household works: Coal combustion generates enormous amount of smoke, soot, dust, CO, $SO_2$, NOx. However, burning of LPG releases fewer amounts of pollutants comparatively.

ii. Gasoline Exhaust: Different categories of vehicles like 2 wheelers, 4 wheelers, heavy duty vehicles, etc. release a number of gaseous air pollutants and particulate matter. They mainly includes NOx, SOx, VOCs, CO, $O_3$, $PM_{10}$, $PM_{2.5}$ and sometimes lead. Vehicles contribute approximately 70% of air pollution as they are major source of primary and secondary air pollutants.

iii. Industries:

   a. Chemical Industries: They generate SOx, NOx, VOCs and PM.
   b. Coal Powered Plants: $SO_2$, CO, NOx and PM.
   c. Electroplating and metallurgical Industries: CO, CO2, NOx, PM, cooper, lead etc.

d. Gasoline-fuel industries: They include petroleum, diesel which emits VOCs, NOx, SOx, CO, PM, $O_3$ etc.
e. Paper manufacturing industries: PM10, PM2.5, $SO_2$ etc.

iv. Agricultural Practices: agriculturally based chemical fertilizers which includes pesticides and herbicides like chlorinated hydrocarbons, etc.

**4. Classes of Main Air Pollutants**

**Table 1**: Summary of air pollutants, sources and impacts

| S.No. | Pollutant | Source/Cause | Effect |
|---|---|---|---|
| 1. | Carbon monoxide (CO) | Vehicular emissions, wood burning, coal burning, incomplete combustion etc. | Nausea, dizziness, severe headache, cardiac arrests, brain stroke, low blood pressure and CO poisoning. After inhalation due to incomplete combustion, CO inhales deep into the lungs and combine with haemoglobin as it has high affinity than oxygen. The resultant is carboxyhaemoglobin. When a large number ofcarboxyhaemoglobin molecules increases in the human body then it decreases the oxygen level. Due to high decrease in O2 levels, the condition becomes fatal and leads to death. |
| 2. | Carbon dioxide (CO2), declared as pollutants only at elevated levels. | Biomass burning | Global warming, greenhouse effect and climate change. |
| 3. | Sulphur dioxide ($SO_2$) | Industrial processes, fossil fuel combustion, wild fire, thermal power plants, smelters and volcanic eruptions | Repiratory ailments, diziness, bronchitis, reduced plant production, yellowing, necrosis and corrosion to marble, spoiling of leather, corrosion. |

| | | | |
|---|---|---|---|
| 4 | Polynuclear Aromatic Hydrocarbons (PAHs) | Gasoline exhaust emissions, emissions from garbage sites, coal tar lining. | Nausea, dizziness, lung cancer, respiratory disorders, eye irritation etc. |

| | | | |
|---|---|---|---|
| 5. | Chlorofluorocarbons (CFCs) | Refrigerators, ACs, cleaning solvents, disinfectants. | Ozone layer depletion, formation of ozone hole. |
| 6. | Nitrogen Oxides | Biomass burning, forest fires, gasoline exhaust emissions, mining etc. | Nausea, dizziness, cardiovascular disorders, photochemical smog, acid rain etc. |
| 8. | Peroxy Acetyl Nitrate (PAN) | Vehicular emissions, Chemical industrial emissions, domestic activities etc. | Irritation of eye, throat and trachea, damages clothing, paints and rubber etc. |
| 9. | Particulate matter (PM 2.5 and PM10) | Vehicular emissions, construction based activities, industrial emissions, wood burning etc. | Respiratory disorders, bronchitis, emphysema, asthma; some are carcinogenic. |

## 5. Air Pollution Control

*A. Source Emissions - Industries*

- Low-sulphur fuels should be encouraged to use to reduce the emission of sulphur dioxide and reduction in benzene in gasoline fuels (petrol) from 10% to 1% is implemented as per central pollution control board norms.
- Exhaust hoods are the latest modifications in industrial ovens which have good efficiency to recover the solvents that have the possibility to become air pollutants.
- Cost effective instruments are implemented to reduce the air pollutant concentrations.
- Removal of pollutants at source reduction like wet scrubber, cyclone separator, electrostatic precipitator etc.

*B. Source Emissions - Vehicles*

- Eradication of Old Vintage Vehicles. The vehicles, particularly cars, which are older than 15 years, should be discarded due to their low efficiency and malfunctioning of engines. These vehicles are also one of the main reasons for emissions of air pollutants in the atmosphere.
- Catalytic converters are efficient devices fitted in engines of different set of vehicles and plays an important role in converting noxious gases in to less harmful gases through chemical reaction.
- Improvement in quality of vehicular fuel which will reduce the tail-pipe emissions and further betters the air quality. For e.g. sulphur content was reduced from 0.5 to 0.2 percent in diesel to reduce the emission of $SO_2$ from tail-pipe. Methyl tert-

butyl ether (MTBE) should be added as a gasoline additive for increasing the octane number and reducing knocking of engine

- Unleaded petrol is recommended to use especially in Delhi to reduce the lead emissions from gasoline fuels.
- Alternative fuels should be used in place of gasoline fuels like CNG, LPG, biodiesel, vegetable oils, ethanol etc to curb the emission of air pollutants..
- To promote the use of electric vehicles

*C. Legal and Policy Measures*

For the control of air pollution, two main acts are implemented, The Air Pollution prevention and Control Act, 1981 and Environmental Protection Act, 1986.

**Implementation of Air Quality Standards (NAAQS)**

- The National Ambient Air Quality Standards (NAAQS) were first suggested under Air Pollution Prevention and Control Act, 1981 for prescribing the stnadrds of different hazardous air pollutants. In 1984, a statuary body called Central Pollution Control Board (CPCB), India adopted 12 parameters which are need to prescribed for air quality standards. The 12 parameters are, particulate matter (PM10 and PM2.5), sulphur dioxide ($SO_2$), nitrogen dioxide ($NO_2$), carbon monoxide (CO), ammonia ($NH_3$), ozone ($O_3$), lead (Pb), benzene ($C_6H_6$), benzopyrene, arsenic and nickel. CPCB had initiated this programme and later called as National Air Monitoring Programme (NAMP).

**National Air Quality Monitoring Programme**

There are about 610 monitoring stations in 227 cities in India as on 1 January 2019. The air quality monitoring was undertaken under the supervision of CPCB, SPCB, DPCC, National Environmental Engineering Research Institute (NEERI), Nagpur and pollution control committees. Meteorological (weather related) parameters is also carried out along with air quality data.

*D. Greenbelt Development/Landscape Green Planning*

- To plant tolerant species which are good absorbers of different air pollutants. These types of plant species are called as sinks and help in purifying the air.
- These tolerant plant species will be screened by a biological method called as Air Pollution Tolerance Index (APTI) and planted around the periphery of the affected area.

*E. Emission Norms and Emission Standards*

Emission norms are the threshold or permissible limit set by nodal agency for a particular category of vehicle. These limits are for those air pollutants which are emitted usually from tail-pipe of vehicles like CO, NOx, VOCs, PM etc. If any vehicle has emission of these air pollutants above the set permissible limit, it is considered as unfit vehicle. The Automobile Research Institute, Pune is the certifying authority for the vehicles fulfilling the standards.

Euro norms are applied in Europe and also for setting up of permissible limits for different categories of vehicles. In India, Euro I norms are referred to as INDIA 2000 because they have been effective since 4 January 2000. The regulations corresponding to Euro II, known as 2005 norms, have not yet been indicated by the Government of India.

*F. Air Pollution Control Equipments*

**i. Control for Particulate Matter**

1. Wet Scrubbers are used to take away the pollutants from furnace flue gas. They consequently pass through scrubbing liquid and hence, particulate matter is removed.
2. Electrostatic Precipitator (ESP) is also a fine device for removal of PM which is based on the principle of opposite charges attract each other. The dust particles which are having positive charge, they get deposited on negative charge electrode (cathode) and similarly, negative charge dust particle deposits on positive charge electrode (anode).
3. Cyclone Separators: Particulate matter is removed by centrifugal force taking place in a rotator machine. It will remove particles 10 microns or larger.

**ii. Control for NOx**

1. Exhaust gas recirculation (EGR).
2. Catalytic converter.

**iii. Systems to Decrease VOC**

1. Gas Flare
2. Biofilters are devices that use living matter to trap biologically degradable pollutants.

## Noise Pollution

Noise is an unwanted and unpleasant sound. Noise pollution may be defined as unwanted sound released into the atmosphere and may create some adverse effects or the unwanted noise. The high intensity sound may have great physical/physiological damaging effects which can be extremely damaging.

The sound intensity has been measured in terms of decibel (dB). The various levels of the sounds with the loudness sensation are given in the Table1.

| S. No. | Noise levels (dB) | Sensation | Source |
|---|---|---|---|
| 1 | 0-20 | Whisper | Rustle of leaves, Soundproof room |
| 2 | 20-40 | Faint | Quite home, Private office, Emptyauditorium |
| 3 | 40-60 | Moderate | Noisy home, conversation quite radio |

| | | | |
|---|---|---|---|
| 4 | 60-80 | Loud | Average radio, average street noise, average factory |
| 5 | 80-100 | Very Loud | Loud street noise, noisy factory, Police whistle |
| 6 | 100-120 | Deafening | Elevated train, Thunder, Nearby riveter |
| 7 | 120-140 | Physical Pain | Jet aircraft, auto horn and Train horn |

**6. Source of sound pollution**

There are number of sources available indoor and outdoor for the sound generation. The most common source in our houses are gadgets like mixture/grinder, vacuum cleaner, washing machines, cooler, air conditioner, radio, TV system, whereas the outdoor sources are factories, vehicles, aeroplane, train, loud speakers and crackers used in the various festivals.

**7. Effects of Noise pollution**

Noise exposure cause variety of adverse health effects and it depends on its intensity, frequency and periodicity.

1. **Effects of high intensity sound:** The sound (80-100dB) of such intensity is emitted by machineries, motor cycles, high speed music system etc. It may cause the emotional and behavioural changes by producing nervous tension and cardiovascular problems like heart disease and blood pressure. Such regular exposure to machineries can cause complete hearing loss, rise in the blood cholesterol level and body plasma concentration.
2. **Effects of Explosive sound:** The sound above 110dB referred as the explosive sound which may be generated by crackers, train, motor cycles etc. Such sound may easily cause vomiting, sever concussions of internal ear and profound deafness.
3. **Effects of loud and sudden noise:** A sonic boom produces startle effects and damage brain. It may also damage the property such as window breaks.
4. **Effects of intermittent noise:** Such sound is very disturbing and may cause of psychiatric illness as found in the area having regular sound of aeroplanes.
5. **Effects of low noise:** This is the common noise produced by crowed, roads, radios and TVs etc. Interfere with the conversation and causes emotional and behavioural stress. Such noise increases the consumption of the alcohol, drug tranquilizers and sleeping pills.
6. **Effects of absolute silence:** Absolute silence is also undesirable for man. An perfect anechoic room (sound levels 0-5dB), one stay for a few minutes, he will be overcome by an uncomfortable sensation as pounding of heart will be felt and ringing noises start in the ears. Mild sound (about 10 dB) found very necessary for the human existence and feel pleasant in such sound produced by nature like bird songs.

### 3. Control of Noise Pollution

Control of noise pollution is basically, reduction of the sound intensity of any available sources. There can be three ways to reduce such type of the noise pollution:

i. Eliminate the noise at the sources,
ii. Modify the path of the sound transmission
iii. Provide receiver with some forms of protection

Some of the important ways for noise reduction are as follows:

1. **Noise pollution reduction in industries:** Noise of the industries can be reduces by replacing old machineries by newer and efficient ones. The noisy generator must be located far from the work area. Factory workers must wear ear-muffs (for sound above 90dB) or ear plug (for sound bellow 90dB). Apart from this some plants such as Ashok, Banayan, Neem, Kadamb etc should be grown around factory to minimise noise pollution.
2. **Reduction of community noise:** The use of loudspeaker in the marriages or other occasions should be banned else permitted for a certain period of time. Measures should be taken to prohibit the manufacture, sale and use of crackers of high sound intensity.
3. **Reduction of traffic noise:** Old and sound producing motor vehicles should be banned on roads and denied operating. Hooting and blowing of horns needless should be restricted legally. All along the highways 50 feet wide plantation strips should be developed.
4. **Reduction of aeroplane and jet noise:** Aerodrome should be located very far from the residential area. The heavy and thick green belts should be developing to reduce the noise pollution levels around aerodrome.
5. **Planning of cities and housing system:** Well-planned cities should be developing to reduce the noise pollution created by the industries and highways. There should be sufficient number of green belts in between residential areas.
6. **Legal control of noise pollution:** Recently, Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCBs) have been given power to the frame certain rule and regulations to control noise pollution. Certain rule and regulations are:
    i. Silence zone should be created near hospitals and educational institute.
    ii. Use of sound amplifiers should be strictly restricted.
    iii. All the vehicle should have effective silencers and uncontrolled blowing of horn should be declared illegal.
    iv. Mid-night aircraft flight should be restricted/minimised.
    v. Restriction on the factory noises should be handled legally and soundless machineries should be promoted.
    vi. Developmental authorities should have a legal procedure and protocol to establish significant number of green belts in and around cities.

# Nuclear Pollution

## 1. What is nuclear hazard: Introduction

Nuclear hazard can be stated as potential risk arising due to exposure to radiation emanating from the atomic nuclei. The important process leading to emission of radiation by different nuclei is known as radioactivity.

The source of radioactivity is emission of energy from radioactive isotopes such as Carbon-14, Uranium-235, Uranium-238, Uranium-239, Radium-226, etc. The energy from these sources is released in the form of alpha, beta or gamma radiation. All these radiations have different impact on tissues. The alpha rays are highly ionizing but have low tissue penetration. If alpha emitting source is ingested it causes severe problem. Beta rays have more penetration ability and can cause cell damage and mutagenesis. It is also used in radiation therapy for cancer patients. Gamma radiations have high penetration value and can cause severe problems. These sources are also used for sterilization of medical and scientific equipment and treatment of food. The nuclear power plants using radioactive elements for energy generation (see **Fig 1.1**) have been among the cause of disasters affecting millions with radiation fallout.

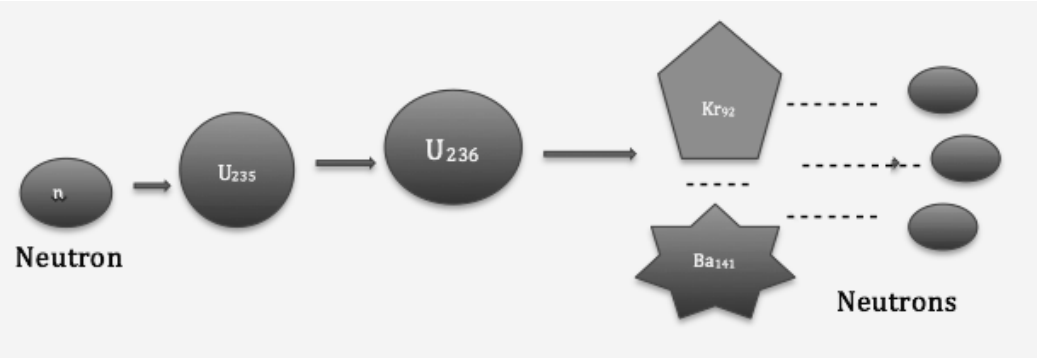



**Figure 1.1**: Nuclear chain reaction occurring in nuclear reactors also known as nuclear fission reaction.

## 2. Causes of Radioactive Pollution

1. **Nuclear accidents:** Nuclear energy is one of the important energy sources discovered in recent time. The high energy in nuclear substances is due to their high latent power, also responsible for high level of radiation. Due to the safety issues, the use of nuclear materials are not promoted much but research is still underway to determine its environmental safety. Nuclear power plant accidents such as the Fukushima Daiichi nuclear disaster, Chernobyl disaster, and Three Mile Island accident are very well-known nuclear disaster that left many dead and many more affected person and contaminated environment by the radiation released.
2. **Weapons of Mass Destruction (WMD):** The Second World War was end up with the use of nuclear missiles and atomic bombs; a form of nuclear energy may easily explain the damaging nature of radioactive pollution. These two strikes in Hiroshima and Nagasaki in 1945 after which the children were reported with complications such

mental retardation, autism and other disorders. In the present situations, the number of cancer cases present in the two towns is more than those of the rest of Japan.

3. **Use of radio isotopes:** These are radioactive isotopes, since they have an unstable atomic nucleus and emit energy and particles when it changes to a more stable form. Each radioisotope has a definite half-life period and characteristic disintegration. Such type of energy liberated mostly in the form of alpha, beta, or gamma rays. Radio isotopes are used to make detectors and in other industrial activities. Isotopes such as uranium have high concentrations of radiation. While the common Isotopes such as carbon containing radioactive material are easily found in water ways through sewage lines. Since most of the raw sewage is untreated before release, once released, the isotope combines with other compounds and elements in water. Consumption of this water through any means is a potential intake of radiation.
4. **Mining:** Mining basically related to the excavation of the mineral ores. Radium and Uranium, for instance, are naturally occurring in the environment and are equally radioactive. Other minerals with a hint of radiation are thorium, plutonium, radon, potassium, carbon and phosphorus.

5. **Tests on radiation:** Due to the lots of interesting properties radiation grabs attention of scientist to carry out research. It plays one of the important roles in the treatment of cancer. Chemotherapy is now widely used in the treatment of cancer in which radiation used to prevent further growth of the cancer cells as well as keep the immune system strong.

### 3. Impact of nuclear hazards

The nuclear hazard can pose a great threat to the environment according to spread of radioactive radiations. These are:

- The radioactive substances are known to be hazardous and when released in environment can disperse or accumulate in living organisms by passing from one trophic level to another in a food chain.
- One of the radioactive substances known as Strontium-90 has properties similar to calcium and can be easily deposited in bones replacing calcium. Contaminated milk is regarded as a source entering human body through ingestion.
- The nuclear power plants use tritium, which has a half-life of 12.3 years, and emit radioactive Beta particles. These power plants are known to be routinely or accidentally releasing tritium into air and water. If tritium enters a body by inhaling or swallowing, the beta particle released by it can bombard cells causing mutation.
- There are certain areas where exposure to nuclear radiation is high and poses threat, which are uranium mineworkers, radium watch dial painters, technical staff at nuclear power plants etc. The people working in these areas are exposed to radiations for longer duration, which can cause cancer, mutations and teratogenesis (prenatal toxicity characterized by structural or functional defects in the developing embryo or foetus) in them.

### 4. Effects of Radioactive Pollution

**a. Short range effects**

The short-range effects are acute and expressed within few days or weeks after the exposure to radiation. The effects may be:

i. physical crippling or

ii. immediate death

**b. Long range effects**

The long-range effects take longer time to express. Such delayed effects of the radiations are now centres of the World's interest. These include:

(i) Genetic changes

(ii) point mutation and chromosomal aberration

(iii) increase incidence of tumour and cancer

(iv) Shortening of the life span

(v) Loss of vitality

(vi) Anaemia

(vii) Haemorrhages etc.

**5. Mitigation of nuclear hazards**

Taking precaution for safe disposal of nuclear waste should be the first priority. The radioactive nuclear waste can be damaging to the environment and so the best way is considered to be burial of wastes in deep trenches.

- **Delay and decay**: The radioactive waste generated from nuclear reactors, industries should be stored in airtight containers and allowed to decay deep beneath the ground in pits.
- **Concentrate and contain**: The small amount of high radioactive waste should be mixed with other components such as concrete and solidified and dumped deep in ocean or beneath the earth.
- **Dilute and disperse**: The moderately or weak radioactive waste should be released in the environment after diluting it with some inert materials.

**Protective Measures**

Some of the protective measures are listed as follows:

1. While handling UV lamps, dark glass spectacles or goggles must be worn. UV rays can't penetrate dark glasses.
2. Visible light neutralises UV damage considerably. Thus exposure to the sun light to any individual exposed to UV could be a good remedial measure to treat the expose individual.

# Water Pollution

Water pollution means one or more substances building up in aquatic bodies to a limit that

they cause trouble life forms. Water pollution also deals with the amount of polluting substance released and alsothe amount of water it is discharged into.

**1. Types of Water Pollution**

There are different perspectives of considering pollution.

***A.*** *Based on water bodies.*

- Surface water pollution – The most evident type of water pollution affects surface waters like huge oceans, lakes, rivers and streams. For example, an oil slick from oil tanker can affect a huge area of the marine region.
- Groundwater pollution – All of Earth's water is not surface water. An immense amount of water is present in the underground of rock structures or aquifers that are not visible to one and people hardly ever think of. Water that is stored in aquifers is groundwater. The rivers and streams are not only nourished by aquifers but they also supply drinking water. Fertilizers applied in the fields often seep into the soil and contaminate the waters. Groundwater pollution though less evident than surface water pollution is also a mounting problem.

***B.*** *Based on source.*

- Point Source – If pollutants are discharged from one place or spot for e.g. an effluent pipe of a plant, oil spillage from a tanker, smoke stcak discharge etc. The pollution from such a source is known as point source pollution.
- Non point source – A prodigious amount of water pollution also happens from variety of speckled sources. This is called non-point source pollution.

**2. Sources of Water Pollution**

**A.** ***Domestic:*** Domestic sewage is wastewater generated from the household activities. It is 99.9 percent pure ater; remaining 0.1 percent are pollutants which are both organic and inorganic materials. Organic materials are food and vegetable waste, excreta, faecal matter whereas inorganic materials such as phosphates and nitrates come from soaps and detergents.

Sewage is an entirely natural matter that should degrade blandly in the environment. Sewage also contains a host of chemical substances, from drugs, papers, plastics, their drainage system. Soaps and detergents used in washing machines and dishwashers are finally drained into the sewage.

**B.** ***Agriculture:*** It is undoubted that extensive use of fertilizers and pesticides, collectively called as agro-chemicals, increases the agricultural output. These, mostly artificial chemicals, move into the waterbodies along with rainfall and groundwater by leaching. Such chemicals are persistent and may get entry into the food chain causing numerous problems in the animals.

**C.** ***Industries:*** Most of the industries are situated along the banks of the river for convenience and disposal of effluents. Often these effluents comprise acids, bases,

dyes, paints, and a range of other chemicals. Detergents form white foam. The industrial waste can be mercury, lead, cadmium, chlorides, fluorides, ammonia etc. Such addition alters the pH of water and turns it into a condition that is fatal to aquatic forms. The industries regularly discharge heavy amount of superheated water used in the thermal power plants, oil refineries, nuclear power plants, etc. such high temperature injuries and kills the aquatic flora and fauna. High temperature also reduces the amount of oxygen dissolved in water that may harm or even kill the aquatic life, the condition is called as thermal pollution.

Global Impacts of Water Pollution

1. **Eutrophication:** It refers to the enrichment of freshwater bodies by accumulation of nitrates and phosphates mostly due to the result of anthropogenic activities. It is predominantly found in sluggish rivers and shallow lakes.

   The number of coastal areas affected by eutrophication globally is over 500. High amounts of nitrates and phosphates are present in sewage, fertilizers and other organic matters. The waters at depth are usually deficient in oxygen, ranging from hypoxic to anoxic condition. A number of algal blooms are identified by discolouration of water bodies often due to the high density of pigmented cells. The water usually becomes greenish. This is known as **'algal bloom'**. Rapid algal expansion leads to increased decomposers. All forms of aquatic life such as decomposers, other aquatic vegetation including the algae, aquatic animals including the fishes, consume the oxygen that is dissolved in the water for respiration. This in turn seeks a great requirement for oxygen and leads to oxygen depletion.

2. **Bioaccumulation and biomagnification: Bioaccumulation** is the accretion of substances like pesticides, metals and various organic compounds into the body of living organism over a period of time. This can occur either because the chemical gains entry faster than it can be utilized, or because the chemical cannot be metabolized by the organism.

Steps to prevent or control water pollution

1. **Awareness:** Creating mass awareness is the foremost step towards preventing and controlling water pollution. Many NGOs have campaigned against over harvesting of fish and pleased for tough penalties against the industries and factories dispensing effluents in the water bodies.
2. **Legislation:** The major problem with water pollution is mainly because of the flowing nature of water that results in trans-boundary pollution. Most of the rivers cross national boundaries whereas ocean and seas may span continents. The pollutants released by the factories in one nation with pitiable emission standards cause glitches in the neighbouring countries. In many a case, the countries that suffer has stringent environmental legislation and elevated environmental standards. The environmental

laws make a situation difficult for the people to foul, but it is more difficult to make such laws practically operative as they need to be implemented across national and international borders. To handle such trans-boundary issues several international negotiations and laws were negotiated and agreed upon:

- The International Convention for the Prevention of Pollution of the Sea from oil tankers, London, 1954.
- The International Convention for the Prevention of pollution from Ships, 1973, (MARPOL 73/78),
- UN Convention on the Law of the Sea, 1982 enforced in 1994.
- London Dumping Convention, 1972.
- Convention on the control of trans-boundary movement of hazardous astes and their disposal, 1989, Basel.
- 1998 OSPAR Convention for the Protection of the Marine Environment of the North East Atlantic.
- Nairobi International Convention on the Removal of Wrecks, 2007

The GOI has enacted the following legislations:

- The Water (Prevention and Control of pollution) Act, 1974 further amended in 1988.
- The Water (Prevention and Control of Pollution) Cess Act, 1977, last amended in 2003.

## Thermal Pollution

The water pollution due to the heat is popularly known as thermal pollution. Hot water from different sources when enters into the water bodies affects the water bodies adversely. Both plant and animals get affected by such type of the pollution.

1. **Sources of thermal pollution:** Several industries, thermal power plants, and nuclear power stations use large quantity of water as a cooling agent, resulting heating of water after the absorption of excess of heat from these processes. In the thermal power plants about 70% of the heat is lost as waste which is absorbed by the cooling water. The water becomes hot and then released into the reservoirs. Such types of water often raise the temperature upto 10-15 degree centigrade and became dangerous for the aquatic ecosystem. Sewage and industrial waste also rise of temperature of water bodies by 4-6 degree centigrade.
2. **Effect of warm water on aquatic life:** Some of the major adverse effects on aquatic life are as follows:
   i. Small animals, phytoplankton, zooplankton, small fishes get affected and killed by the hot water.
   ii. Cell walls of such type of animals and plants severely damage due to the hot water.
   iii. Hot water may adversely affect the enzyme and metabolic activities of the plants and animals.
   iv. The quantity of the dissolve oxygen also get effected by the presence of the hot water as the oxygen present into the water body bobbles out from the water body. It is also called as the gas bobble diseases of water. This results increase in the BOD levels of the water and causes suffocation of aquatic animals.
   v. The hot water makes the aquatic organism susceptible towards toxic chemical and bacterial/viral/fungal infections.
   vi. Thermal pollution adversely affects the overall ecological balance of the aquatic ecosystem, e.g. it affects the hatching of the eggs of some fishes, while other insects begin hatching with slight rise in the water temperature.
3. **Control of Thermal Pollution:** It is very hard to control thermal pollution because it is the results of actions of many people. But in case of already known source location, the water should be allow to cool down before discharged. The cooling towers and fountains can be used for the rapid cooling. Large tank and reservoir should be constructed to retain the water for a little longer time. It is also suggested to discharge such type of the hot water system into the running water bodies instead of stagnant water bodies.

## Soil Pollution

Soil is a natural medium of inorganic and organic nutrients and has an inbuilt system of spontaneous recycling of matter. It is affected by changes in the atmospheric conditions as well as water contents and microbial population.

Soil pollution is defined as an undesirable change in the natural, physical, chemical or biological components of the soil.

4. **Sources of Soil Pollution:** The various sources of soil pollution may be categorized into two groups:
    1. **Natural Sources:** They are as follows:
        - **i. Plant Residues:** Normally plants, on death and decay, contribute organic matter to the soil and thereby increase soil fertility. Sometimes residues from crops, fields and orchards carry plant pathogens and pests. Burning of crops yields residues with CO (8.3%), NO (1.5%), hydrocarbons (5,3%) and particulate matter (8.5%).
        - **ii. Animal Residues:** Animal wastes, such as faecal matter, urnine, blood, slaughter house waste in the form of liquid or particulate matter, bodies of dead animals etc. are all indiscriminately dumped into the soil. Excessive organic contents are harmful for healthy growth of roots as they create hypertonic conditions in the soil causing wilting or stunted growth of the plants.
    2. **Anthropogenic Sources**
        - **i. Domestic:** Domestic sewage is wastewater generated from the household activities. It is 99.9 percent pure ater; remaining 0.1 percent are pollutants which are both organic and inorganic materials. Organic materials are food and vegetable waste, excreta, faecal matter whereas inorganic materials such as phosphates and nitrates come from soaps and detergents.
        - **ii. Agriculture:** It is undoubted that extensive use of fertilizers and pesticides, collectively called as agro-chemicals, increases the agricultural output. These, mostly artificial chemicals, move into the waterbodies alongwith rainfall and groundwater by leaching. Such chemicals are persistent and may after gain entry into the food chain causing numerous problems in the animals.
        - **iii. Industries:** Most of the industries are situated along the banks of the river for convenience and disposal of effluents. Often these effluents comprise acids, bases, dyes, paints, and a range of other chemicals. Detergents form white foam. The industrial waste can be mercury, lead, cadmium, chlorides, fluorides, ammonia etc. Such addition alters the pH of water and turns it into a condition that is fatal to aquatic forms. The industries regularly discharge heavy amount of superheated water used in the thermal power plants, oil refineries, nuclear power plants, etc. such high temperature injuries and kills the aquatic flora and fauna. High temperature also reduces the amount of oxygen dissolved in water that may harm or even kill the aquatic life, the condition is called as thermal pollution.
        - **iv. Radioactive Waste:** High concentrations can be lethal, whereas low concentrations of radioactive substances can cause malignancy and various forms of sickness. Two most important contributors in Europe are Sellafield and Ca La Hague.

**2. Impacts of Soil pollution**

i. Soil pollution decreases soil fertility. Loss of soil fertility makes the soil inhabitable for plants and other organisms.

ii. Soil pollution has adverse effects on soil microbial population that plays important role in soil formation and keeping it fertile.

iii. Indiscriminate use of agro-chemicals destroys the soil flora and fauna and soil properties.

iv. $SO_2$, $SO_3$ and oxides of nitrogen of the atmosphere are washed down by rain in the form of $H_2SO_4$, $HNO_2$ and $HNO_3$ which reach the soil and increase the acidity. Increased acidity develops unhealthy conditions for plant growth.

v. Dumping of radioactive wastes onto the soil is lethal to plants and microbial life. Such pollution has far reaching effects and even human beings are not spared.

**3. Control measures for Soil pollution**

1. Control of pesticide pollution can be carried out by:
   i. Lowering its use in agriculture.
   ii. Use of degradable insecticides like organic phosphates.
   iii. Use of short-lived chemical pesticides.
   iv. Adopting biological control
   v. Releasing sterile males into the natural population of insects.
   vi. Rotation and diversification of crops etc.
2. Control of fertilizer pollution
3. Control of pollution due to garbage
4. Control of acid rains
5. Control of radioactive based soil pollution
6. Discharge of hot liquids and oil leakage in the soil should be avoided.
7. Sewage should be biologically treated and converted into manure instead of its uncontrolled release into the soil.